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DEVELOPMENT RESEARCH

# Machine learning-based data processing for predictive modeling in mechanical systems

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**Abstract**
Machine learning (ML) has become the prevalent practice in the field of predictive modeling in mechanical systems, which allows the identification of performance patterns and detection of early signs of a malfunction. The use of MS together with conventional mechanical approaches like finite element evaluation has improved structural analysis and failure assessment. Furthermore, the ML has revealed fresh sets of material properties enabling the creation of new and unique materials since ML scans a vast database to uncover new potentialities. When we have appropriate specifications for mechanical material and apply ML algorithms to the design process, the correct combinations of elements and configurations are revealed. The paper also highlights the difficulties encountered when dealing with big and tremendously large data, the significance of preprocessing in training and how the various categories of training Learning; Supervised, Unsupervised and Reinforcement learning apply to material design. This work offers a great overview of the current use of ML in analyzing mechanical systems and material science and presents a possible trajectory for further development of the predictive modeling and material design.

**Keywords** Predictive modeling · Data preprocessing · Machine learning · Material design · Feature extraction

## 1 Introduction

It is now possible for dynamic mechanical systems to be designed with predictive modeling as the key driving force in means of enhancing on reliability, performance and some even on Resource utilization rates. Data preprocessing is an important step of the pre-processing phase of predictive modeling where raw data is to be pre-processed with different techniques that include cleaning, transforming and feature extraction for use in predictive modelling. Data preprocessing is significant to have clean, relevant, organized, and standard dataset for modeling. They include missing value handling, feature scaling and selection. According to Bishop (2006a) noise reduction through preprocessing was also said to enhance model accuracy. In the same respect, Montgomery and Runger (2014) described statistical approaches for managing intricate-engineering data. Most computations in mechanical systems are applied in series preprocessing techniques when it comes to time-series data, vibrations, and other various sensor-based inputs. Furukawa and Natsume (2020) emphasized preprocessing techniques for predictive maintenance and Sharma and Aggarwal (2021) presented how feature engineering enhances the identification of mechanical failures. Feature extraction is the process of dimensionality reduction with relative retention of important information, which makes it a basis of predictive analytics. In a survey of feature extraction techniques for mechanical systems, Schmidt et al. (2019) considered the capability of reducing computing requirements. More recently, Bhattacharjee and Datta (2022) reported on other techniques of feature extraction, highlighting their relevance to fault detection and characterization, as well as dynamic system identification. These studies stress the fact that accuracy of the extracted features defines the predictive potential of mechanical models in question. ML has made a major difference in the field of modelling and prediction of mechanical systems due to its application of data analysis. Alpaydin (2020) made a point to focus on the effects of increase in preprocessing and in

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increase in the performance of ML model. ML was integrated in Yin and Kaynak (2015) with conventional mechanical practice including finite element analysis to enhance structural analysis and fault prediction. Further, Chen and Yu (2018) used preprocessing vibration and signal data to capture the improved fault diagnosis performance that leads to an enhanced classification. Sundararajan and Adams (2020) highlighted how ML techniques improve reliability when paired with pre-processed data. These studies confirm that robust data processing is essential for early fault detection and system optimization.

In mechanical systems, materials are the building blocks of our civilization; from the tools we use to the structures we build. Over time, our understanding of the link between a material's structure and its properties (mechanical, thermal, optical, etc.) has accelerated the discovery of new materials. Among these properties, mechanical properties are crucial. They determine a material's strength, durability, and resistance to issues like wear, cracking, or bending (Meyers and Chawla 2008).

Designing mechanical materials involves fine-tuning their composition and structure to achieve specific properties. This is significant for all sorts of new generational materials, ranging from smart synthetic and architecture composites to biomimetic materials with optimized natural like architectures (Barbero 2017). Metamaterials—a new type of materials having certain properties, which are different from their composite materials, are also at the frontier. But since these advanced materials possess elaborate requirements, they offer enormous design opportunities, requiring fresh strategies (Goh et al. 2022).

Several challenges exist in developing and deploying smart sensors; however, artificial intelligence, especially ML, has been identified as a suitable solution to these difficulties (Winston 1992). Since these algorithms can consider lots of datasets to make their decisions, they are designed suitable to solve complicated issues in materials science and design. As we would see in traditional approaches of ML, scientists still must define features which are simply representations of data. However, the developments in this area are sobering, although the recent method of deep learning (DL), which is an integral part of ML, has changed this greatly. DL goes through and employs a mixture of complicated neural network structures for the purpose of 'learning' representations of the needed data. This has led to advances that are beginning to transform fields such as computer vision, natural language processing and self-driving cars.

Noting that the field of materials design can greatly benefit from ML, researchers have been recently employing it more often, as can be seen by the rising number of publications and reviews. ML-based materials research typically involves three key components: well-aligned material data set, model choice and concise research question. They pointed out that data preprocessing is crucial. Here information about raw material is converted to numerical form or descriptors which can be input to the chosen model of ML. This step entails both material science and ML skills in order to provide the correct and time bound training.

This work discusses the current state of mechanical design using ML and prescribes a general approach to research in the area. We will discuss what kind of data is collectible or separately generable for the purpose of ML models, document previous uses of ML in this field, and touch upon approaches to data acquisition and preparation, data preprocessing, problem framing, and selection of the ML models. Such an approach has the potential to become a game-changer in mechanical design and discovery through uniting mechanics, the science of materials and ML.

This paper aims at discussing how ML has impacted the predictive modeling in mechanical systems through enhancing the systems performance, availability, and fault identification. It includes an extended analysis of combining ML with conventional mechanical engineering approaches, including finite element analysis, for better structures prediction. Secondly, the efficient strategy for preparing challenging mechanical system datasets is described. The paper presents the use of feature extraction, noise reduction, and time series data analysis as fundamental underlying methods and illustrate how they can affect the accuracy and performance of ML on mechanical applications. Thirdly, the role of using ML and its subfield, namely deep learning, in the search of mechanical materials and optimization of these materials is discussed. It describes how ML algorithms approach the identification of material compositions that matter the most, process large design spaces efficiently, and originate cutting-edge abstract materials including metamaterials and bioinspired structures. Finally, the paper presents several potential challenges, for example, working with big and complex data and ML supplementation of traditional mechanical approach, and demonstrates how future work can address these. It outlines the areas of innovation, predictive modeling, materials science and the design of efficient sustainable mechanical systems.

Section one of this paper defines the topic and presents a literature review on the application of ML for predictive modeling and materials design. Section 2 details ML techniques, including supervised, unsupervised, and reinforcement learning, applied to material design. Section 3 examines data processing methods such as feature extraction and noise reduction. Section 4 discusses strategies for creating high-quality datasets tailored for material design with ML. Section 5 focuses on preparing data for ML, including descriptor generation and preprocessing. Finally, Sect. 6

concludes with key insights and future directions in ML-driven material design and predictive modeling.

## 2 Machine learning techniques for material design

ML offers a powerful toolkit for designing new materials. Just like learning styles differ among people, there are various approaches within ML. Here's a breakdown of some key methods (Schmidt et al. 2019; Yu et al. 2023):

- **Supervised Learning:** This method is like learning with a teacher. It uses labeled data (data with predefined answers) to train algorithms to map inputs to desired outputs. Common algorithms include linear regression, decision trees, and gradient boosting. These are great for tasks like predicting material properties or classifying materials into specific categories.
- **Unsupervised Learning:** Unlike supervised learning, this approach is more like exploring on your own. It analyzes unlabeled data to identify hidden patterns and relationships within the data itself.
- **Reinforcement Learning:** Imagine learning through trial and error. This method involves an agent interacting with an environment and receiving rewards for making good decisions. The goal is to maximize these rewards over time (Abdallah and Sahnoun 2022).

There's also semi-supervised learning, which combines labeled and unlabeled data for training, offering a middle ground between supervised and unsupervised approaches.

For material design, supervised learning is a popular choice due to its reliability and well-established techniques. However, the field is not limited to these classical methods.

Materials and process design have indicated numerous approaches detailed in the following Table 1. These approaches include several methods such as the ML technique, the Bayesian technique and the generative model's technique. These methods are useful to the researcher to determine favourable microstructures, associate processing stages to properties, and forecast properties. Some of the

**Table 1** Key approaches in materials and process design using machine learning techniques

| Category | Approach | Focus | Techniques/tools | Examples |
|---|---|---|---|---|
| Materials Design | Microstructure Sensitive Design (MSD) | Identifying microstructures for desired properties | Texture evolution modeling, processing streamlines | Crystallographic texture optimization (Li et al. 2007; Fullwood et al. 2010; Adams et al. 2012) |
| | | | Genetic Algorithms | Optimizing texture for specific properties |
| | | | Surrogate-Based Optimization | Handling uncertainties |
| | | | Generative Models | Generating microstructures with continuous variables |
| | | | | Microstructure reconstruction (Bompas and Sandfeld 2024; Wijaya et al. 2024) |
| Process Design | ML-Based Process Optimization | Linking processing steps with material properties | PCA, neural networks | Deformation process modeling (Acar and Sundararaghavan 2016, 2018; Sundar and Sundararaghavan 2020; Lin et al. 2023) |
| | Database-Driven Approaches | Mapping microstructures to processing paths | Databases, autoencoders | Microstructure-path mapping (Sundararaghavan and Zabaras 2005) |
| | Bayesian Approaches | Solving inverse process-property problems | Gaussian processes, active learning | Estimation of process parameters (Generale et al. 2024, 2023; Tran et al. 2020; Morand et al. 2022) |
| Combined Design | Texture Evolution Networks | Process-structure–property linkage | Directed graphs, graph search | Optimal paths for microstructure evolution (Shaffer et al. 2010) |
| | Knowledge Graphs | Descriptor-based property prediction | Graph neural networks | Property prediction (Thomas et al. 2023; Kumar et al. 2020b) |

trends for the future study in this field are construct of FAIR-compliant data platforms; multiscale modeling; digital twin; artificial intelligence experiment, and AI ethic for designing and developing materials.

### 2.1 Deep learning: unveiling complexities

Artificial neural networks (ANNs), especially deep learning (DL) models, have become game-changers in ML. Think of them as complex networks of interconnected units inspired by the human brain. Some prominent architectures include:

- **Feedforward Neural Networks (FFNNs):** These process information layer by layer, like passing a message through a team (Bhattacharjee and Datta 2019; Cao and Zhang 2022).
- **Convolutional Neural Networks (CNNs):** Imagine analyzing images. CNNs excel at this by extracting key features through specific operations (Rai et al. 2020; Alpaydin 2020).
- **Recurrent Neural Networks (RNNs):** Useful for sequential data like music or language, RNNs consider past information to predict future outcomes (Yin and Kaynak 2015; Chen and Yu 2018).

### 2.2 Generative models: creating new possibilities

These models, like generative adversarial networks (GANs) and variational autoencoders (VAEs), can even create entirely new data points based on existing patterns (Sundararajan and Adams 2020). Imagine creating new materials with desired properties!

- **Generative adversarial networks (GANs):** Think of two rivals—a forger (generator) and a detective (discriminator). GANs pit these against each other, with the generator creating realistic-looking data and the discriminator trying to identify fakes. This competition refines the generation process.
- **Variational autoencoders (VAEs):** Imagine compressing information into a smaller form and then recreating it. VAEs do this with data, encoding it into a simplified form (latent space) before decoding it back into a reconstruction.

### 2.3 Enhancing the toolbox: advanced techniques

ML is not just about creating models; it's also about improving them. Here are some methods for fine-tuning ML performance:

- **Bayesian Learning and Gaussian Process Regression (GPR):** These techniques help with parameter estimation and uncertainty measurement, leading to more robust models.
- **Active Learning:** This method involves strategically selecting data points for labeling, allowing the model to focus its learning on areas that will benefit it most.

Emerging Techniques: Graph Neural Networks.

A recent development is graphing neural networks (GNNs), particularly graph convolutional networks (GCNs). These are adept at learning from data with connected structures (like graphs) and show promise for various material science and mechanics applications (Meyers and Chawla 2008).

## 3 Data processing in material design

Just like cars need fuel to run, ML models need data to function effectively. High-quality data is especially crucial for material design using ML. But how much data do we really need? Where can we get it? And most importantly, how good is the data we have, and how can we improve it? These questions are fundamental because the data acts as a bridge between the material design problems we're trying to solve and the ML models we use to solve them.

There are two main ways researchers get data:

a) **Existing Sources:** This involves collecting data from scientific publications, databases, or online repositories.
b) **Homegrown Data:** This involves generating new data through experiments or computer simulations.

However, simply throwing data at an ML model is not enough. There are challenges:

### 3.1 Data availability

- **Data availability:** If obtaining data is too difficult or expensive, using ML might not be the best approach. For instance, if there's already a reliable method to explore all possible material designs, there's no need for ML.
- **Data Coverage:** If the data only covers a limited range of materials, the ML model's usefulness is restricted.
- **Data Understandability:** Images or text data might be clear to us, but machines can't understand them directly.

This is where data preprocessing comes in. It involves cleaning and transforming raw data into a form that ML models can understand. Here, a researcher's expertise in materials science is vital. They can select the right data, ensure it represents the problem well, and preprocess it effectively for optimal ML model performance (Barbero 2017; Goh et al. 2022; Winston 1992).

Choosing the right ML method is crucial for success in material design. Table 3 unveils various ML tools and their functionalities. Each method is explained in a clear, concise way, highlighting its key characteristics. A new "Benefits" column showcases the advantages of using each method for material design tasks. For instance, linear regression is praised for its simplicity and ability to capture linear relationships, while CNNs excel at analysing image and voxel-based data, making them ideal for capturing spatial relationships in materials.

Table 2 also provides practical examples, referencing relevant citations, to illustrate how each ML method is being applied in mechanical materials design. This comprehensive overview empowers researchers to make informed decisions when selecting the most suitable ML tool for their specific material design challenges.

## 4 Creating high-quality data for material design with ML

While running experiments or simulations offers flexibility in data creation, there's a balancing act. Researchers need to consider the cost of generating data versus the resulting model performance. Studies show that incorporating knowledge of materials science and mechanics leads to more representative datasets and better models.

Here's how computational methods come in:

- **Finite element analysis (FEA):** This method simulates materials at a larger scale, like analyzing 3D microstructures and their properties.
- **Molecular dynamics (MD):** This method zooms in on the atomic level, simulating the behavior of individual atoms within a material.

To reduce the computational cost of these methods, researchers use data-driven approaches with reduced-order models. Additionally, symmetries in materials (like repeating patterns) can help reduce the design space that needs exploration. For example, imagine designing a composite material with a repeating tile structure under symmetrical loading conditions. We only need to analyze a representative tile instead of the entire structure.

Luckily, there are resources to help researchers compare and improve their ML models. Benchmark databases, like Mechanical MNIST, allow researchers to compare the accuracy and efficiency of different ML techniques. Once a model is trained, high-throughput experiments can validate its predictions. In some cases, active learning can be used to identify areas where more data is needed. Here, the model itself suggests which additional experiments would be most beneficial.

Finally, there's even a system that can automate some of this process! Imagine a system that uses a technique called Bayesian optimization to decide which experiments to run next. This could be a game-changer for tackling complex challenges in mechanical materials design, like optimizing structures for additive manufacturing.

## 5 Getting data ready for machine learning in material design

Just like your car needs gas to run smoothly, ML models need data to function effectively. But before feeding data into a model, it often needs some cleaning and prep work, a process called preprocessing. This ensures the data is in the right format for the model to understand and use it efficiently (Goodfellow et al. 2014).

There are a few key techniques used in data preprocessing for material design with ML:

- **Data Augmentation:** Imagine having a small dataset of images. Data augmentation helps create more data by flipping, rotating, or slightly modifying the existing images. This gives the model more information to learn from.
- **Filtering Irrelevant Data:** Not all data points are equally important. Sometimes, irrelevant details can confuse the model. Filtering removes these unnecessary details and helps the model focus on what truly matters.

Here are some real-world examples of data preprocessing in action:

- **Predicting Cracks in Materials:** Researchers studying how cracks form in brittle materials used data from simulations. They transformed the complex atomic structure information into a simple image format. This allowed the model to focus on the spatial features of the cracks, ignoring unnecessary details about individual atoms.
- **Building a Model from Existing Data:** Another study used data on glass materials collected from various sources. A technique called least angle regression helped identify the most important features from this large dataset. This ensured the model learned from the most relevant information.
- **Making the Most of Small Datasets:** Sometimes, collecting a large amount of data can be expensive or time-consuming. In such cases, researchers might use image processing techniques like rescaling or cropping to adjust the existing data and make it more usable for training a complex deep learning model.

**Table 2** Demystifying machine learning tools for material design

| ML Method | Explanation | Benefits | Example Applications in Mechanical Materials Design |
|---|---|---|---|
| Linear Regression/Polynomial Regression | Fits a straight line (linear) or a curved line (polynomial) to relate input data to an output variable | Simple to understand and interpret<br>Good for capturing linear or polynomial relationships between variables | Predicting material properties like modulus (Goodfellow et al. 2016) and strength (Mitchell 1999) |
| Support Vector Machine (SVM)/Support Vector Regression (SVR) | Creates a hyperplane (decision boundary) to separate data points in high-dimensional space | Effective for high-dimensional data classification and regression<br>Can handle complex, non-linear relationships | Predicting material properties like strength (Mitchell 1999) and hardness (Bishop 2006b)<br>Structural topology optimization |
| Random Forest | Combines multiple decision trees, where each tree makes a prediction, and the final output is based on the majority vote | Robust to outliers and overfitting<br>Handles complex datasets well | Predicting material properties like modulus (Russell and Norvig 2009) and toughness (Russell and Norvig 2003) |
| Feedforward Neural Network (FFNN) | Connects artificial neurons (nodes) in layers, with information flowing in one direction | Flexible and can learn complex relationships from data<br>Widely applicable for various prediction tasks | Predicting various properties like modulus, strength, toughness, and hardness (Haykin 1999)<br>Design of spinodoid metamaterials (Fukushima and Miyake 1982) |
| Convolutional Neural Networks (CNNs) | Analyzes data with a grid-like structure (images or 3D voxels) using convolutional filters to extract features | Excellent for image and voxel-based data analysis<br>Effective at capturing spatial relationships in materials | Predicting mechanical properties like strain fields, elastic properties, stress fields, and yield strength (LeCun et al. 1999) |
| Recurrent Neural Network (RNN)/Long Short-Term Memory (LSTM)/Gated Recurrent Unit (GRU) | Processes sequential data by feeding information back into the network, allowing it to consider past information | Well-suited for analyzing sequential data like time series or fracture patterns<br>Can model complex temporal relationships | Predicting fracture patterns, plastic behaviors, and multi-scale modeling of porous media (Hochreiter 1998) |
| Generative Adversarial Networks (GANs) | Pit two neural networks against each other: a generator that creates new data and a discriminator that tries to identify fakes | Can generate new data points based on existing data distributions<br>Useful for exploring new material design possibilities | Predicting property distributions, stress/strain fields, composite design, and architected materials (Hochreiter and Schmidhuber 1997) |
| Gaussian Process Regression (GPR)/Bayesian Learning | Treats model parameters as random variables and estimates their probability distribution | Provides uncertainty quantification for model predictions<br>Useful for problems with limited data or high uncertainties | Predicting material properties like modulus and strength [22, 23, 24<br>Design of super compressible metamaterials |
| Active Learning | Selectively queries a user or external source to label new data points, focusing on areas that improve model performance | Reduces the need for large, pre-labeled datasets<br>Improves model efficiency by focusing on valuable data points | Predicting material properties like strength |
| Genetic or Evolutionary Algorithms | Mimics biological evolution principles to optimize a desired objective function | Well-suited for problems with complex search spaces.—Can find innovative solutions that traditional methods might miss | Predicting material properties like hardness |

**Table 3** Machine learning for mechanical material design: a powerful toolbox

| Area | Descriptions | Examples |
|---|---|---|
| Property Prediction | Train ML models to predict mechanical properties (like Young's modulus, strength, hardness) based on material data (composition, structure) | Predicting fracture toughness of polycrystalline silicon using Random Forests (RF) and Feedforward Neural Networks (FFNN) |
| Material Discovery | Leverage ML to identify materials with superior properties beyond those in the training data | Developing high-entropy alloys with superior hardness using Support Vector Regression (SVR) |
| Microstructure Analysis | Analyze complex microstructures using Convolutional Neural Networks (CNNs) to understand their influence on mechanical properties | Predicting the effective elastic properties of composites using 3D CNNs analyzing microstructure images |
| Fracture Prediction | Train ML models to predict how cracks propagate in materials, aiding in designing materials with enhanced crack resistance | Using Convolutional LSTMs to predict fracture patterns in crystalline solids based on atomistic simulations |
| Nonlinear Material Behavior | Develop ML models to handle complex material behaviors like plasticity, fracture, and dynamic impact | Employing Recurrent Neural Networks (RNNs) to predict the stress–strain behavior of elasto-plastic composites |
| Topological Design | Design innovative material structures with superior performance using ML models and optimization algorithms | Optimizing strength and fracture toughness of composite materials using CNNs and genetic algorithms |
| Data-Driven Computational Methods | Integrate ML with computational mechanics to tackle problems where governing equations are unknown but large datasets exist | Developing a data-driven framework using neural networks for modeling and designing new composite materials |
| Multi-scale Modeling | Utilize RNNs to link models at different scales, enabling the prediction of material behavior across various length scales | Using RNNs to connect smaller-scale simulations with larger-scale models for porous media |
| Microstructure Characterization | Employ unsupervised learning techniques to automatically identify and classify microstructural features from images | Leveraging unsupervised learning to identify and characterize grains in polycrystalline solids |
| Solving Partial Differential Equations | Train deep neural networks to solve complex partial differential equations governing mechanical behaviors | Using deep neural networks to solve various solid mechanics problems |

One of the appealing attributes that belong to the specific category of DL models is that they can take less amount of preprocessing than such ML methods as, for instance, SVM or LR. This is because in DL models the training process itself can identify and extract important features of the raw data being used. Furthermore, the work on the characterization of data-driven solvers may extend ideas for the fast handling of noise and/or missing data which is sometimes faced in material response investigations.

In Table 3, the author clearly describes an idea of how ML is revolutionizing mechanical material design. Which demonstrates several use cases within different phases of the design process. It can consequently forecast key material properties, search for completely new materials that outperform those currently in use and analyse microstructures to determine how they affect mechanics. It demonstrates how solutions to problems such as fracturing crack prediction or numerous material properties are approached in a general sense using ML techniques. This work also discusses the application of ML to enable the researcher to create new structures of materials and its compatibility with other methods of calculations. While the examples presented above and shown in Table 2 illustrate the applicability of ML to MMD, it is important to stress how and in what aspects ML has proven to be an incredibly flexible toolbox for shedding light and providing solutions to open research problems; from multi-scale modeling to fully automating microstructure analysis.

## 6 Conclusion

Artificial intelligence (AI) has made a definitive shift in the field of materials and mechanical engineering research, improving data deluge through employing technology of ML in materials designing. Categorized methods, for example, legacy methods and tool-based approaches, generally depend on the researcher's judgment on embracing new materials, which inhibits discovery. On the other hand, ML is more capable for dealing with more massive data volume and involves both the material characteristic and the mechanical factors right from the get-go thus allowing researchers to consider a much wider array of new materials possibilities.

New exciting developments in physics constrained ML frameworks are extending even more the applicability of these technologies. The works, including the hybrids of the physic laws with ML models like physics informed neural networks (PINNs), show that integrating physical laws with the ML models could be useful for increasing accuracy and might also provide fresh approaches to materials design. Not only do these models forecast the traits of materials, but they also provide scientists with the ability to discover additional principals governing the interaction of materials, thus providing us with further understandings of material characteristics. Such approaches as GNNs are also providing fresh analytical solutions in providing a better fundamental understanding of how materials function and integrate.

There is evidence of material reciprocity, where materials science benefits from ML while, at the same time, the latter relies on the former. In result, when ML techniques address issues in materials design, they will also inform how insights achieved from materials research can enhance ML algorithms. This symbiotic interaction improves the progression of novel machine learning algorithms particular to structure–property relationships in materials design, which includes hierarchical structures and multifunction material characteristics.

Nevertheless, some issues linger, primarily, regarding how the resulting ML models arrive at their decisions. Most of the ML algorithms are 'black box' deterministic, and this makes it really hard to understand how they came to a certain conclusion. With our knowledge of materials science, we can better understand these algorithms and improve the interpretability of models that consider the complex topological characteristics of new metamaterials.

The capability that ML brings to the future of material design is promising. Though challenges remain and need further discussion, integration of ML with materials science holds the promise to dramatically change how we will design, discover, and model new materials to innovate a myriad of industries.

## Declarations

**Conflict of Interest** The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author declares the following financial interests/personal relationships which may be considered as potential competing interests: None.

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